

Syed Fuzail Farhan

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fuzail127.github.io/fuzzymuseum

linkedin.com/in/fuzail-farhan

Mechatronics engineering student focused on robot integration, motor control, and mechanical prototyping, with hands-on experience connecting sensors, firmware, and physical hardware.

Education

University of Waterloo - BAsC, Mechatronics Engineering (Co-op)

Experience

Research Assistant, IDEAS Clinic - University of Waterloo

May 2026 - Aug 2026

- Integrated a Dobot arm, Orbbec depth camera, and Python/OpenCV pipeline to identify and sort coloured parts for the Toyota Innovation Challenge; added YOLO hand detection to pause motion while a hand occupied the work zone.
- Co-authored an AGMA-versus-FEA spur-gear study; investigated mesh convergence and rebuilt the ANSYS Mechanical contact definition, reducing contact-stress error against AGMA from 76.2% to 5.13%.
- Worked through rapid R&D cycles to meet project deadlines, integrating, testing, and troubleshooting robotic hardware and software for engineering challenges.

Research Assistant, IDEAS Clinic - University of Waterloo

Aug 2025 - Dec 2025

- Designed SolidWorks models and 3D-printed components for experimental validation; developed a heat-exchanger test assembly with printed adaptors, a TPU buffer, and an O-ring seal for a co-authored Ansys publication.
- Co-designed an ANSYS FEA/CFD workshop for 70+ students and developed a microcontroller/FPGA workshop for 60+ students covering PWM, serial communication, and oscilloscope-based circuit debugging.

Computation Team Co-op, FormulaNano

Jan 2025 - Apr 2025

- Developed Python pipelines and extended an in-house quantum chemistry codebase to automate UV-Vis and NMR spectra generation, replacing manual steps with unattended batch processing.

Software Engineering Intern, GrazeAI

May 2024 - Aug 2024

- Built a reusable Python scraper using REST APIs and regex to filter 600,000+ clinical trials, reducing manual trial identification from days to minutes.

Design Teams

Firmware Developer, UW Robotics Team

Jun 2026 - Present

- Contributing to the migration of rover arm-to-base communications to Ethernet between two NVIDIA Jetsons and an STM32 controller.

Firmware Engineer, Electrium Design Team

Sep 2024 - Dec 2024

- Implemented three-phase BLDC field-oriented control for an ESP32 vehicle speed controller; configured DRV8353 gate drivers and tuned dead time and drive strength with an oscilloscope.

Selected Projects

EMG-Operated Claw

In progress

- Integrated analog EMG sensing, ESP32 processing, and servo control in a 3D-printed claw; exploring TPU geometries and electrode placement for more consistent readings.

Maze-Navigation Robot

In progress

- Built a battery-powered ESP32 robot with DRV8833 motor drivers, PWM-controlled DC motors, and HC-SR04 ultrasonic sensing; developing multi-robot coordination.

Technical Skills

Mechanical: SolidWorks, AutoCAD, GD&T, ANSYS Mechanical/Discovery, FEA/CFD, FDM 3D printing

Embedded and Test: STM32 (HAL), ESP32 (ESP-IDF), Arduino, ADC, PWM, UART, SPI, I2C, oscilloscopes, multimeters

Software: C, C++, Python, MATLAB, OpenCV, Git, Linux

Portfolio

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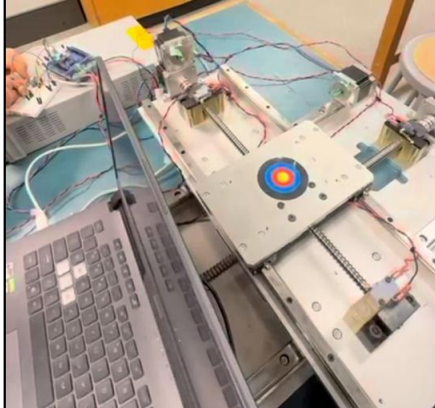
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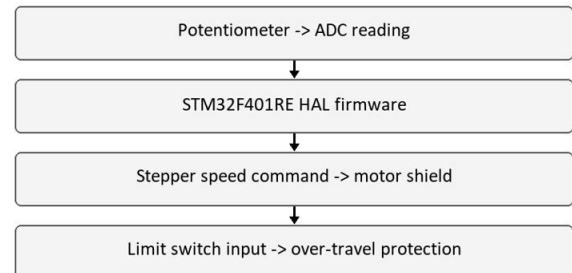
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Two Axis Motor Control

Completed lab project | STM32F401RE, HAL, Embedded C



Two-axis positioning apparatus



Firmware signal flow

- Extended an existing HAL codebase to read a potentiometer through the ADC and map its input to stepper motor speed.
- Implemented polling and interrupt-driven GPIO synchronization and limit-switch logic to prevent mechanical over-travel.
- Completed the lab implementation by connecting analog speed input, firmware control, motor actuation, and end-of-travel handling within the existing codebase.

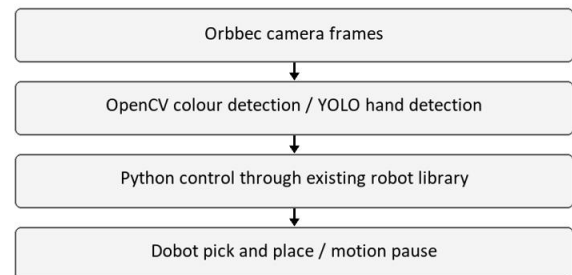
Project details

Vision Guided Pick and Place

IDEAS Clinic | Toyota Innovation Challenge | May - Aug 2026



Dobot test cell



Implemented processing and control flow

- Integrated a Dobot arm and Orbbec depth camera with Python/OpenCV colour segmentation to pick coloured objects and place them in a container, using the lab's existing robot-control library.
- Added a YOLO hand-detection condition that pauses motion while a hand is detected in the work zone; tested variations in object colour and pick/drop locations.
- Built and tested challenge solutions to support students. The hand-detection feature is a prototype behavior, not a validated safety system.

Project details | [Challenge repository](#)

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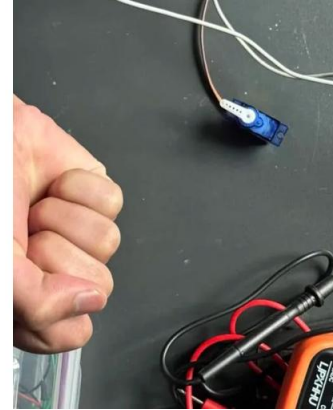
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EMG Operated Claw

Personal project | ESP32, analog EMG sensor, servo actuation | In progress



Printed claw geometry



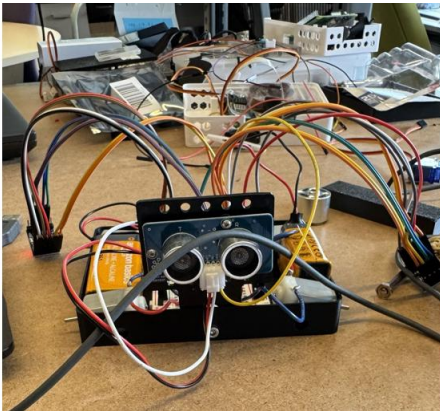
EMG and servo experiment

- Processed muscle-activation signals from an analog EMG sensor on an ESP32 and used threshold logic to command a servo controlling a 3D-printed claw.
- Exploring TPU claw geometries and electrode placement to improve consistency. Repeatability remains a development target rather than a measured result.
- Combined a printed mechanism, sensor input, embedded processing, and servo actuation in one prototype; ongoing work focuses on more consistent operation.

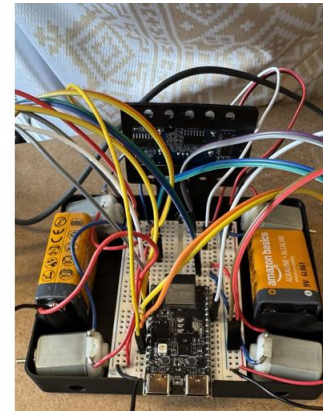
Project details

Autonomous Maze Solver

Personal project | ESP32, Embedded C, ultrasonic sensing | In progress



Robot prototypes with ultrasonic sensors



Breadboarded motor-control electronics

- Built a battery-powered robot prototype around an ESP32, DRV8833 dual H-bridge motor drivers, and brushed DC motors; used PWM to control motor speed.
- Integrated HC-SR04 ultrasonic sensing for wall-distance and obstacle detection, providing inputs for wall-following and dead-end navigation.
- Developing the platform toward coordinated maze exploration with multiple robots. Shared mapping and fleet coordination remain in progress.

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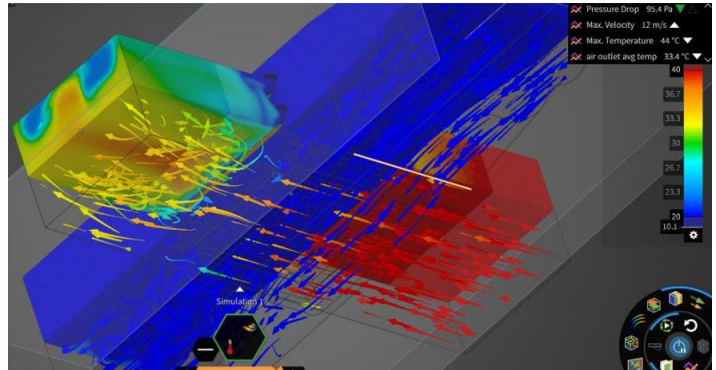
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Printed Heat Exchanger and Experimental Validation

IDEAS Clinic | Co-authored with Andi Zhao | Aug - Dec 2025



Cross-flow test assembly



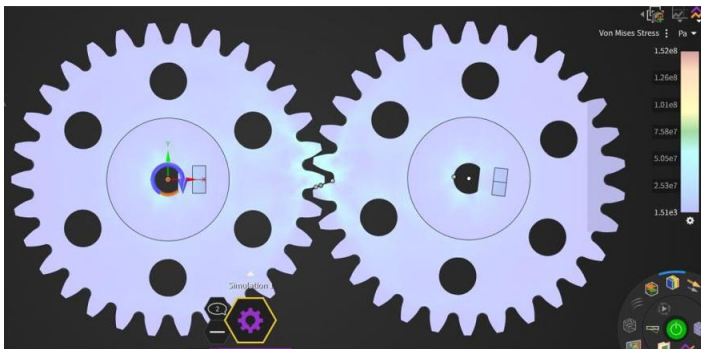
ANSYS flow simulation

- Developed a copper-filled HTPLA heat-exchanger test assembly with printed adaptors, a TPU buffer, O-ring seal, and flow restrictor; measured flow and temperatures for simulation inputs.
- Used heated specimens, a thermal camera, and transient simulation to estimate filament conductivity near 0.7 W/(m K). Investigated porosity, sensor placement, and geometry differences.
- Re-measuring outlet airflow addressed backpressure in the assembled rig. The study explains remaining model/experiment differences and was published as an Ansys education resource.

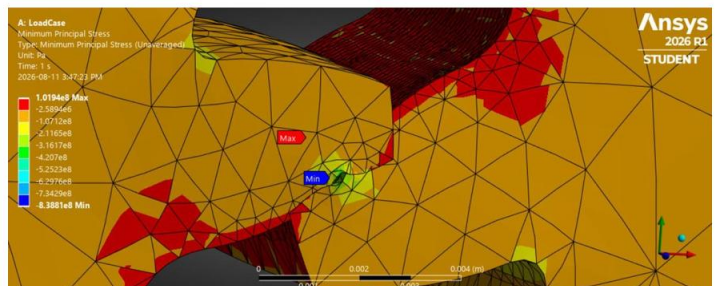
Full case study | Ansys publication

Spur Gear Stress Analysis

IDEAS Clinic | Co-authored with Owen Grimm and Jessica Liu | May - Aug 2026



30-tooth gear pair and applied loading



Refined contact mesh

- Compared AGMA calculations with ANSYS FEA for steel gears with 2 mm module, 10 mm face width, and a 32 N m applied moment; examined root bending and contact stress.
- Discovery predicted 210 MPa contact stress against 883.3 MPa analytically. Mesh refinement did not resolve the discrepancy; rebuilding the contact definition in Mechanical produced 838 MPa, or 5.13% error.
- Documented the tradeoff: the final root-stress error was 6.93%. The case study explains model assumptions and limitations and is under Ansys review.

Full case study